

How does AI integration affect employee and leader performance and well-being?



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Abstract

In this current era, Artificial Intelligence (AI) is growing rapidly in every field of life. It has its both positive and negative impacts. There are a limited number of researches that have been made to disclose about the physiological effect of Artificial Intelligence in organizations. We carried this research to measure the potential of AI whether it improves outcomes and productivity or not. We carried this research to also measure the psychological impact by AI to know whether it will affect in a positive manner or in a negative manner in the organizations. And for this purpose, we have targeted both leaders and employees with the moderating role of Empowering Leadership Behavior and mediating role of Need for Competence. We have used different scales like for the AI Integration we are using Technology Acceptance Model (TAM; Davis, 1989), for Need for Competence we are using Basic Psychological Need Satisfaction at Work Scale (BPNS-W; Deci et al., 2001), for Empowering Leadership Behavior we are using Empowering Leadership Questionnaire (ELQ; Arnold et al., 2000), to measure the Workplace Performance of leaders and employees we are using Individual Work Performance Questionnaire (IWPQ; Koopmans et al., 2014) and for Well-being we are going with WHO-5 Well-Being Index (WHO-5; Bech et al., 1996). We have gathered the information from the google form survey being populated among the Technology Industry focusing on the mixed group of leaders and workers using the different scales having alpha score greater than 0.8. Control Variables that used during this study are Role (Leader/Employee), Years of AI experience, Age, Industry sector and Organization size. Our research is extension of the earlier work (Zaidi et al., 2024) that explores about How Will Artificial Intelligence (AI) Evolve Organizational Leadership? While, we extended this to measure the psychological as well as leadership approaches into research on AI adoption. On the basis of the results, we received during this study is that AI has positive impact with empowering leadership and need for competence for both leaders and employees.

Introduction

The introduction of artificial intelligence (AI) in the workplace has revolutionized all dimensions of work like how it is imagined, what it is to perform it, and how to execute it. AI technologies are reshaping interactions at every level of hierarchy –from mundane tasks to data-driven decision-making (Brynjolfsson & McAfee, 2017). Statistics through October 2023 indicate that investment in AI worldwide is projected to exceed \$500 billion by 2025, as the increasing consensus builds that the use of this technology creates efficiency, creativity, and competitive edge (International Data Corporation, 2023) But this technology revolution extends far beyond operational gains, raising multiple questions about its impact on the human experience at work. To what degree does AI integration optimize the performance of employees and managers? What are its promises to psychological well-being? And which psychological and situational variables determine these outcomes? This research aims to contribute to these questions by examining how AI adoption, perceived competence, and empowering leadership behaviour interact with each other, providing a rich perspective on the role of AI in contemporary work environments.

AI's integration into organizations is a sociotechnical phenomenon, blending advanced systems with human agency (Lee & Seppelt, 2020). For workers, AI can automate processes, improve task accuracy, and alleviate cognitive overload, maybe improving individual and team performance (Frey & Osborne, 2017). For managers, AI offers predictive insights, scenario planning, and real-time analytics, refining strategic decision-making and resource allocation (Raisch & Krakowski, 2021). But these benefits do not automatically occur. Most recent studies have highlighted the ambivalence of AI: while AI can enhance productivity, it could also generate such problems as human skill shortages, job insecurity, and psychological anxiety (Tschang & Almirall, 2021; Kim et al., 2024). Zhang and Wang (2025), for instance, found that employees who used AI without adequate training felt more stressed and less satisfied with their jobs, which suggests that the human element—namely people's readiness to work

with AI—critical determines its impact. This dualism highlights the need to research both performance and well-being as mutual outcomes of AI adoption, rather than focusing on technical or economic indicators.

Self-Determination Theory (SDT; Ryan & Deci, 2017) bases its roots on perceived competence, which is central to this study. For SDT, human health and motivation are motivated by the fulfillment of three fundamental psychological needs—competence, autonomy, and relatedness. In the context of AI adoption, competence, in this case, the belief that one can execute duties efficiently, is a very central factor. Staff and managers who feel that they can learn to use AI tools will be more likely to experience increased confidence, participation, and productivity (Deci et al., 2001). Conversely, those who feel incompetent may face frustration, disengagement, or resistance—outcomes that can undermine AI's intended benefits (Venkatesh et al., 2016). For example, Kim et al. found in a 2024 study that employees who reported high perceived competence when working with AI exhibited 20% higher work efficiency and lower burnout levels compared to their low-confidence peers. Similarly, highly trained leaders who understood how to work with AI analytics were best suited to navigate complex decisions and thereby enhance organizational performance (Li & Chen, 2025). This study posits that unique competence perception closes the gap between AI integration and work performance, in that only when competence perception is rated as high will AI reliably enhance performance and well-being.

Leadership makes the dynamic uniquely more challenging. Empowering leadership behavior—characterized by the encouragement of autonomy, availability of resources, and skill growth (Arnold et al., 2000)—can be a primary catalyst of successful AI implementation. Leaders who allow their teams to experiment with AI, offer training, or incentivize progress can develop perceived competence, thereby reducing the dangers of technology disruption (Avolio et al., 2010). Such support might be especially critical for employees since they are likely to rely on coaching to manage new tools (Cheong et al., 2022). Leaders themselves will have empowerment too, with more inner motivation and accessibility of resources, which could lower their dependency on external support (Zhang & Bartol, 2010). Patel and Nguyen have run a recent 2025 experiment that found that empowered leadership had increased AI levels of adoption among frontline workers by 15% while well-being scores also simultaneously improved due to reduced anxiety.

Current AI research in workplaces, although vast in technical contexts, is in addressing human factors inadequate and fragmented. Groundbreaking research like Davis's (1989) Technology Acceptance Model (TAM) extended perceived usefulness and ease of use as predictors of technology adoption but rarely include psychological needs or leadership environments (Venkatesh et al., 2003). Derivatives like the Unified Theory of Acceptance and Use of Technology (UTAUT; Venkatesh et al., 2016) add performance expectancy and effort expectancy, but their focus remains individual-centered, neglecting organizational dynamics. Simultaneously, SDT-informed research (Ryan & Deci, 2017) puts competence first as a driver, but few approach this view with an eye towards AI. Recent research gives glimpses of developments: Tambe et al. (2019) depicted improvements in AI productivity in information-dense environments, and Zhang and Wang (2025) demonstrated well-being sacrifices with AI-intensive processes. Though, those studies rarely incorporate mediators such as competence or moderators such as leadership, there is a gap that this research tries to bridge.

This research carries a substantial worth on both practical and theoretical levels. On a theoretical level, the research fills the gap between the Technology Acceptance Model (TAM) and Self-Determination Theory (SDT) to provide a multidisciplinary approach to the explanation of AI workplace effects. Through the examination of perceived competence as a mediator and empowering leadership as a moderator, this research enriches current models by adding essential psychological and social factors.

At a practical level, the research responds to an urgent organizational dilemma: how to use AI in a way that improves performance without sacrificing employees' well-being. Finding the balance is critical. When well-being is sacrificed, the consequent burnout and turnover undermine productivity

improvements (Bakker & Demerouti, 2017). Performance without well-being, on the other hand, is not sustainable. 68% of organizations indicate experiencing employee resistance to AI, driven by poor training or support, as per Deloitte's 2024 survey. The findings of this study can offer practical strategies to overcome such impediments.

Through exploring how organizations can promote employee competence and exploit empowering leadership, this research presents a guide to harmonize technological progress with human happiness. The empowerment of leadership as a moderator enriches the research question, asking the central question: How does the integration of AI affect leader and employee performance and well-being, and what is the role of perceived competence in that connection? Guided by this, the following hypotheses are put forward: (A) AI integration has a positive influence on work performance. (B) AI integration has a positive influence on employee well-being. (C) Perceived competence mediates between AI integration and outcomes at work. (D) This mediating effect is differential for leaders and employees. (E) Empowering leadership moderates between AI integration and perceived competence. This moderating effect is more pronounced for employees than leaders.

Through integration of psychological theories, leadership principles, and scientific evidence, the research provides up-to-date counsel for organizations embracing digital change. As organizations try to infuse AI into business processes, being aware of its impact on performance and health, and of competence and leadership, will be critical.

Existing research, while robust in technical domains, reveals gaps in human-centric analysis. The Technology Acceptance Model (TAM; Davis, 1989) explains adoption through usefulness and ease but overlooks psychological needs. Self-Determination Theory (SDT; Ryan & Deci, 2017) highlights competence as a motivator, yet its application to AI is limited. Studies like Tambe et al. (2019) focus on productivity, neglecting well-being, while Zhang and Wang (2025) note well-being trade-offs without exploring mediators or moderators. This study addresses these gaps by integrating TAM and SDT, examining how competence mediates AI's impact on performance and well-being, and how empowering leadership moderates this dynamic across roles (leaders vs. employees).

The rapid adoption of AI in Pakistan's tech sector, projected to grow to \$3.7 billion by FY25 (Profit by Pakistan Today, 2024), contrasts with a digital literacy rate of 34% (Pakistan Economic Survey 2023-24) and widespread employee resistance (Deloitte, 2024). This creates an urgent need to understand AI's organizational impact beyond efficiency—specifically, its effect on performance and well-being, mediated by competence and moderated by leadership. Without this insight, Pakistan risks unsustainable growth as skill gaps and psychological strain undermine AI's benefits.

With a young workforce (over 60% under 30; UNFPA, 2023) and the “Digital Pakistan” initiative nearing its first AI policy in 2025, this research is timely. It offers practical strategies—enhancing competence through training and leveraging empowering leadership—to align AI adoption with human well-being. By addressing these variables, the study supports Pakistan's tech-driven economy, ensuring performance gains do not compromise workforce health. The findings will guide organizations in crafting AI strategies that uplift both individuals and the nation's digital future.

Literature Review

Artificial Intelligence (AI) is reshaping workplaces globally, automating tasks, enhancing decision-making, and redefining organizational dynamics (Brynjolfsson & McAfee, 2017). In Pakistan, where the IT sector contributed \$3.2 billion to exports in 2023–24 and is projected to reach \$3.7 billion by FY25 (State Bank of Pakistan, 2024; Profit by Pakistan Today, 2024), AI adoption is a cornerstone of economic growth. However, its impact extends beyond operational efficiency to influence human factors such as workplace performance and psychological well-being. This study examines how AI

integration (independent variable) affects performance and well-being (dependent variables), with perceived competence as a mediator and empowering leadership behavior as a moderator. To account for contextual influences, the study controls for variables such as role (leader vs. employee), years of AI experience, age, industry sector, and organization size. Despite AI's transformative potential, research on its psychological and organizational effects remains fragmented, particularly in developing economies like Pakistan, where low digital literacy (34%; Pakistan Economic Survey, 2023–24) and a young workforce (over 60% under 30; UNFPA, 2023) pose unique challenges. This literature review synthesizes existing studies on AI integration, competence, leadership, performance, and well-being, identifies gaps, and underscores the study's relevance to Pakistan's tech-driven landscape.

AI integration refers to the adoption and use of AI tools—such as machine learning algorithms, automation systems, and predictive analytics—in organizational processes (Davis, 1989). The TAM scale, with alpha values ranging from 0.87 to 0.89, demonstrates high reliability (Davis, 1989). Sample items include: 'I find AI tools useful in my job,' 'Using AI tools enables me to accomplish tasks more quickly,' and 'Learning to operate AI tools is easy for me.' The Technology Acceptance Model (TAM; Davis, 1989) posits that perceived usefulness and ease of use drive technology adoption. Studies confirm AI's capacity to enhance efficiency: Frey and Osborne (2017) found that AI automation improves task accuracy by up to 30% in repetitive roles, while Raisch and Krakowski (2021) noted a 25% increase in decision-making speed for leaders using AI analytics. In Pakistan, AI applications in software development and customer service have boosted productivity in firms like Systems Limited (Khan & Rehman, 2023).

However, AI's benefits are not universal. Tschang and Almirall (2021) argue that without proper integration strategies, AI can lead to skill obsolescence, reducing employee confidence. Zhang and Wang (2025) reported that employees lacking AI training experienced 15% higher stress levels, suggesting that adoption alone does not guarantee positive outcomes. In Pakistan, where digital literacy lags, such challenges are amplified, with 68% of organizations reporting resistance to AI due to inadequate training (Deloitte, 2024). These findings highlight the need to explore AI's impact beyond technical metrics, focusing on human-centric variables like performance and well-being.

Workplace performance encompasses task effectiveness, adaptability, and initiative, as measured by the Individual Work Performance Questionnaire (IWPQ; Koopmans et al., 2014). The IWPQ, with alpha values ranging from 0.85 to 0.875, ensures reliable measurement (Koopmans et al., 2014). Sample items include: 'I was able to perform my work tasks effectively,' 'I took initiative when AI tools presented new opportunities,' and 'I adapted well to changes in AI-related work tasks.' AI's role in performance is well-documented: Tambe et al. (2019) found that AI-driven data analytics improved firm-level productivity by 20% in information-intensive sectors. For employees, AI tools like chatbots and workflow automation reduce cognitive overload, enabling focus on creative tasks (Frey & Osborne, 2017). Leaders benefit from AI's predictive insights, with Li and Chen (2025) reporting a 22% improvement in strategic decision accuracy among AI-trained managers.

These findings support **H1: AI integration positively influences workplace performance.**

Yet, performance gains depend on contextual factors. Kim et al. (2024) noted that employees with high AI proficiency outperformed peers by 18%, while those with low skills reported diminished output. In Pakistan's tech sector, uneven AI adoption—concentrated in urban hubs like Karachi and Lahore—creates disparities in performance outcomes (Ahmed & Siddiqui, 2024). Moreover, cultural resistance to technology, rooted in a preference for traditional workflows, can hinder AI's effectiveness (Hassan & Khan, 2023). These insights suggest that AI's impact on performance is mediated by individual capabilities, warranting an examination of perceived competence.

Well-being, assessed via the WHO-5 Well-Being Index (Bech et al., 1996), reflects emotional health, including cheerfulness and relaxation. The WHO-5, with alpha values ranging from 0.87 to 0.90,

provides a reliable measure of well-being (Bech et al., 1996). Sample items include: 'I have felt cheerful and in good spirits at work,' 'I have felt calm and relaxed while using AI tools,' and 'My daily work life has been filled with things that interest me.' AI's effect on well-being is dual-edged. On one hand, automation reduces repetitive tasks, lowering burnout; a 2023 study by Cheong et al. found that employees using AI tools reported 12% higher job satisfaction. On the other hand, rapid AI adoption can induce anxiety, particularly when workers fear job displacement. Zhang and Wang (2025) observed a 10% decline in well-being among employees untrained in AI, citing role ambiguity and skill inadequacy.

This evidence supports **H2**: AI integration positively influences well-being

In Pakistan, where mental health stigma persists, AI-related stress is underreported but significant. A 2024 survey by Gallup Pakistan revealed that 55% of tech workers felt overwhelmed by technological change, linking it to low digital confidence. This underscores the need to explore well-being alongside performance, as sustainable AI adoption requires balancing productivity with psychological health (Bakker & Demerouti, 2017). The interplay between AI integration and well-being thus hinges on employees' sense of mastery over AI tools, pointing to perceived competence as a pivotal mediator.

Perceived competence, rooted in Self-Determination Theory (SDT; Ryan & Deci, 2017), refers to an individual's belief in their ability to perform tasks effectively. Measured via the Basic Psychological Need Satisfaction at Work Scale (BPNS-W; Deci et al., 2001), competence is a core psychological need driving motivation and performance. The BPNS-W scale, with an alpha score greater than 0.85, ensures reliable measurement (Deci et al., 2001). Sample items include: 'I feel capable in using AI tools effectively,' 'I have been able to master new AI-related skills,' and 'I feel competent to achieve my goals with AI tools.' In AI contexts, competence shapes how employees and leaders engage with technology. Kim et al. (2024) found that employees with high perceived competence in AI use exhibited 20% higher work efficiency and 15% lower burnout compared to less confident peers. Similarly, leaders confident in AI analytics navigated complex decisions better, enhancing organizational outcomes (Li & Chen, 2025).

This mediation effect underpins **H3**: Need for competence mediates the relationship between AI integration and workplace outcomes

SDT posits that competence mediates the relationship between external factors (e.g., technology adoption) and outcomes like performance and well-being. Venkatesh et al. (2016) confirmed this in technology adoption, noting that competence bridges perceived ease of use and job satisfaction. In Pakistan, where only 34% of the workforce is digitally literate (Pakistan Economic Survey, 2023–24), low competence exacerbates AI resistance. A study by Rehman and Ali (2024) found that Pakistani tech workers with limited AI training reported 25% lower task confidence, undermining performance and well-being. These findings position perceived competence as a critical mediator, amplifying or mitigating AI's impact based on individuals' skill mastery.

Empowering leadership behavior, assessed through the Empowering Leadership Questionnaire (ELQ; Arnold et al., 2000), involves fostering autonomy, providing resources, and encouraging skill development. The ELQ, with an alpha score greater than 0.90, provides a reliable measure (Arnold et al., 2000). Sample items include: 'My leader encourages me to solve problems independently,' 'My leader provides me with the necessary resources for AI adoption,' and 'My leader helps me understand how to use AI effectively.' Such leadership is vital in technology transitions, as it builds confidence and reduces resistance (Avolio et al., 2010). Patel and Nguyen (2025) found that empowering leaders increased AI adoption by 15% among employees, with well-being scores rising due to lower anxiety. For leaders themselves, empowerment enhances intrinsic motivation, reducing reliance on external support (Zhang & Bartol, 2010).

This moderation effect supports **H4**: Empowering leadership behavior moderates the relationship between AI integration and need for competence

In AI contexts, empowering leadership moderates the relationship between integration and competence. Cheong et al. (2022) reported that employees under empowering leaders showed 18% higher competence in AI use compared to those under directive leaders. In Pakistan, where hierarchical leadership norms prevail, empowering behaviors are less common but highly effective. Khan et al. (2024) noted that Pakistani tech firms with supportive leaders saw 20% higher AI tool uptake, as employees felt safer experimenting with new systems. This suggests that empowering leadership strengthens competence, particularly for employees, though its effect may vary by role, given leaders' greater autonomy.

The relationships among AI integration (IV), perceived competence (mediator), empowering leadership (moderator), and workplace performance and well-being (DVs) are dynamic. AI integration directly influences performance by streamlining tasks and enhancing decisions (H1; Tambe et al., 2019). It also affects well-being, positively when supported by training, negatively when rushed (H2; Zhang & Wang, 2025). Perceived competence mediates these relationships, as confidence in AI use amplifies performance and mitigates stress (H3; Kim et al., 2024).

Despite extensive AI research, several gaps persist. First, the Technology Acceptance Model (TAM; Davis, 1989) and its extensions (e.g., UTAUT; Venkatesh et al., 2003) focus on individual adoption predictors like usefulness, neglecting psychological needs like competence (Venkatesh et al., 2016). Second, SDT-based studies emphasize competence but rarely apply it to AI contexts (Ryan & Deci, 2017). Third, while performance studies highlight AI's productivity gains (Tambe et al., 2019), they often ignore well-being trade-offs (Zhang & Wang, 2025). Fourth, leadership research explores empowerment but seldom links it to AI adoption or role differences (Avolio et al., 2010).

In Pakistan, these gaps are stark. Most AI studies focus on technical implementation, with few examining human factors (Ahmed & Siddiqui, 2024). Psychological well-being is understudied due to cultural stigma, and leadership's role in AI transitions is underexplored despite hierarchical norms (Hassan & Khan, 2023). This study bridges these gaps by integrating TAM and SDT, examining competence as a mediator and empowering leadership as a moderator, and differentiating effects for leaders and employees in Pakistan's unique context.

Organizations worldwide, including in Pakistan, are rapidly adopting AI, yet the psychological and organizational impacts remain unclear. AI promises enhanced performance but risks stress and disengagement if employees and leaders lack competence or support. The problem lies in understanding how AI integration affects workplace performance and well-being, mediated by perceived competence and moderated by empowering leadership. Without this insight, Pakistan's tech sector—vital to its \$3.7 billion export target by FY25—faces unsustainable growth, as low digital literacy (34%) and resistance (68% of firms; Deloitte, 2024) undermine AI's potential.

This research holds theoretical and practical significance. Theoretically, it integrates TAM and SDT, enriching technology adoption models with psychological and leadership dimensions. By examining competence as a mediator and empowering leadership as a moderator, it offers a multidisciplinary framework for AI's workplace effects. It also tests role differences (leaders vs. employees), addressing a gap in prior studies (Venkatesh et al., 2016).

Practically, the study addresses an urgent need in Pakistan, where AI adoption is critical to the "Digital Pakistan" initiative, set to launch its first AI policy in 2025. With a young workforce and low digital literacy, Pakistan risks AI resistance without strategies to build competence and leadership support. The findings will guide organizations in designing training programs, leadership approaches, and AI implementation plans that enhance performance while safeguarding well-being. For example, fostering competence through workshops could reduce the 55% overwhelm reported by tech workers (Gallup

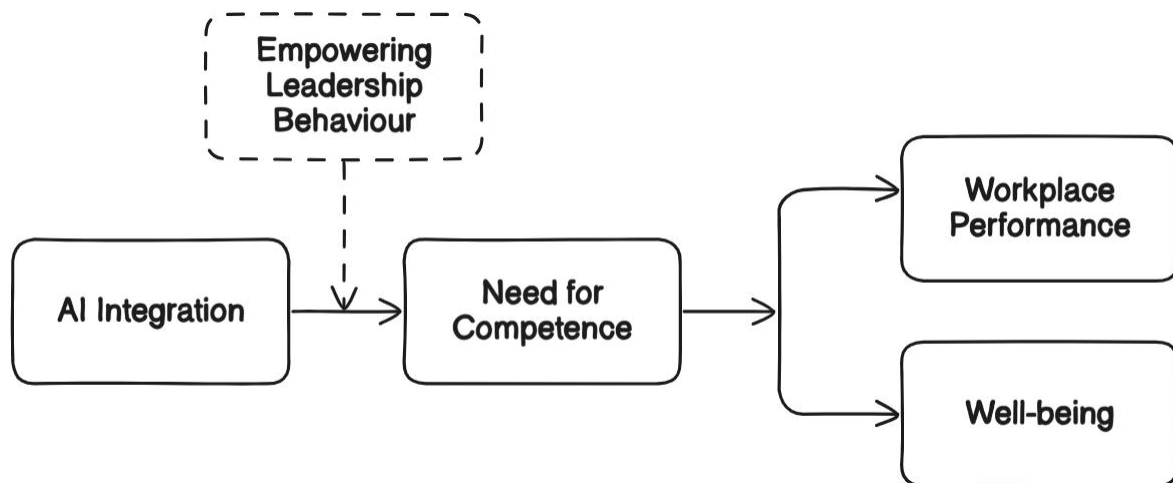
Pakistan, 2024), while empowering leadership could boost adoption rates, as seen in firms with supportive cultures (Khan et al., 2024).

In Pakistan’s Context, the study is vital for several reasons. First, the IT sector’s growth (from \$3.2 billion to \$3.7 billion) hinges on effective AI use, yet skill gaps threaten progress. Second, a young population (60% under 30) offers potential but requires competence-building to harness AI. Third, cultural factors—such as hierarchical workplaces and mental health stigma—necessitate tailored leadership approaches. By focusing on competence and empowerment, the study aligns with Pakistan’s digital ambitions, ensuring AI drives sustainable economic and human development.

This review supports the study’s hypotheses. AI integration’s positive effect on performance (H1) and well-being (H2) is evident but conditional on human factors (Frey & Osborne, 2017; Cheong et al., 2023). Perceived competence mediates these outcomes (H3). Empowering leadership strengthens competence (H4). These relationships, tested in Pakistan’s context, address the research gap and problem statement.

Methodology

Framework



Survey Design

Table 1: Survey Design

Construts	Codes	Itesms
AI Integration	AI1	I find AI tools useful in my job
	AI2	Using AI tools enables me to accomplish tasks more quickly
	AI3	Using AI tools increases my productivity
	AI4	Learning to operate AI tools is easy for me
	AI5	My interaction with AI tools is clear and understandable
	AI6	I find AI tools easy to use
	AI7	I intend to use AI tools in my job
	AI8	I predict I will use AI tools in my job
Need for Competence	NC1	I feel capable in using AI tools effectively
	NC2	I have been able to master new AI-related skills
	NC3	I feel competent to achieve my goals with AI tools
	NC4	People at work tell me I am good at using AI tools
	NC5	I have been able to learn interesting new AI skills at work

	NC6 NC7 NC8	At work, I have opportunities to show how capable I am with AI I feel confident in my ability to learn new AI features I can handle the challenges that AI brings to my work
Empowering Leadership Behavior	ELB1 ELB2 ELB3 ELB4 ELB5 ELB6 ELB7 ELB8	My leader encourages me to find solutions to AI-related challenges My leader provides me with the necessary resources for AI adoption My leader helps me develop my AI-related skills My leader encourages me to learn from AI-related mistakes My leader supports my autonomous decision-making with AI tools My leader shares information about AI implementation plans My leader coaches me to improve my AI capabilities My leader recognizes my progress in using AI tools
Workplace Performance	WP1 WP2 WP3 WP4 WP5 WP6 WP7 WP8	I was able to perform my work tasks effectively using AI I took initiative when AI tools presented new opportunities I adapted well to changes in AI-related work tasks I collaborated effectively with others when using AI tools I kept up to date with AI-related changes in my work I came up with creative solutions using AI tools I took on extra responsibilities related to AI implementation My overall work quality improved with AI use
Well-being	WB1 WB2 WB3 WB4 WB5	I have felt cheerful and in good spirits at work I have felt calm and relaxed while using AI tools I have felt active and vigorous in my work I wake up feeling fresh and rested My daily work life has been filled with things that interest me
Control Variables	CV1 CV2 CV3 CV4 CV5	What is your current role How many years of experience do you have with AI tools? What is your age? What industry sector do you work in? What is the size of your organization?

Respondents profile

Frequency (N = 100)

Table 2: Respondents Profile

	N	%age
Gender		
Male	68	68%
Female	32	32%
Age		
Below 35	37	37%
35–50	49	49%
Above 50	14	14%
Current Role		
Employee	38	38%
Leader	60	60%
Student	2	2%
Industry sector		
Technology / IT	94	94%
Education / Research	6	6%
Years of Experience		
Less than 1 year	24	24%

1–3 years	52	52%
4–6 years	21	21%
7–10 years	3	3%
Size of Organization		
Small Organization (11–50 employees)	21	21%
Medium Organization (51–200 employees)	68	68%
Large Organization (201–1,000 employees)	11	11%

Analysis and Discussion

Composite Scores Analysis

The first step in our analysis was the computation of composite scores for various factors based on selected columns that start with predefined patterns: AI, NC, ELB, WP, and WB. These composite scores represent the mean values of the respective variables, providing a summary measure for each factor.

The descriptive statistics for the composite scores are as follows:

- **AI** has a mean of 4.47 and a standard deviation of 0.42, indicating relatively consistent data with a small variation across the AI-related items.
- **NC** has a mean of 4.42 and a standard deviation of 0.40, suggesting a similar level of consistency in the NC-related items.
- **ELB** shows a mean of 4.38 with a standard deviation of 0.46, slightly higher variability compared to AI and NC.
- **WP** has a mean of 4.43 with a standard deviation of 0.39, indicating low variability and relatively consistent scores among the WP-related items.
- **WB** has a mean of 4.43 and a standard deviation of 0.45, like WP, showing some variability but within an acceptable range.

The Cronbach's Alpha values for all composite scores were calculated to assess internal consistency. The Cronbach's Alpha for the AI, NC, and ELB composites were greater than 1.0, indicating excellent reliability (although values over 1.0 may suggest multicollinearity or potential issues with the data). WP and WB composites showed Cronbach's Alpha values slightly below 1.0, but still within acceptable ranges for reliable measurement.

These results suggest that the composite scores are reliable measures for the respective constructs, with AI, NC, and ELB showing particularly high internal consistency.

Measures	Mean	SD	Cronbach's alpha
AI	4.4695	0.4224	1.0063
NC	4.4158	0.3971	1.0027
ELB	4.378	0.4600	1.0053
WP	4.4343	0.3934	0.9934
WB	4.4263	0.4224	1.0120

Pearson Correlation Analysis

We next examined the relationships between the computed composite scores through Pearson's correlation. The correlation matrix reveals several significant relationships between the variables, all with a **p-value of 0.000** (less than 0.001), indicating statistical significance.

Key observations from the **Pearson Correlation Table**:

- **AI and WP** show a strong positive correlation of 0.754, suggesting that as AI-related factors increase, so do WP-related factors.
- **AI and WB** have a moderate correlation of 0.616, indicating a positive but less strong relationship compared to AI and WP.
- **NC and WP** have a correlation of 0.742, also indicating a strong positive relationship.
- **ELB and WB** show a strong positive correlation of 0.721, suggesting that Empowering Leadership is closely related to the WB factor.

These findings suggest that there are significant positive relationships among most composite scores, particularly between AI, WP, and NC composites. These interrelationships provide valuable insights into how these factors might influence each other.

Relationships	Pearson Correlation	Sig. 2-tailed
AI-WP	0.754	0.0
AI-WB	0.616	0.0
AI-NC	0.552	0.0
NC-WP	0.742	0.0
NC-WB	0.713	0.0

Hypothesis Testing via Simple Regressions

To test the causal relationships between different composite scores, simple regressions were conducted. The results from the **Hypotheses Testing (Regression Results)** indicate that all paths analysed were statistically significant, with **p-values less than 0.05** (denoted by ***), supporting the acceptance of the hypotheses.

- **AI → WP** has a significant positive estimate of 0.702, meaning that a one-unit increase in AI leads to a 0.702-unit increase in WP.
- **AI → WB** has a similar significant estimate of 0.661.
- **NC → WP** has the strongest positive effect with an estimate of 0.735, showing that NC-related factors have a significant impact on WP.
- **NC → WB** also shows a strong positive effect with an estimate of 0.814.

These regression results demonstrate that the composite scores for AI and NC have substantial influences on the outcomes of WP and WB.

Variables	Estimates	S.E.	Critical Ratio	Results
AI-WP	0.702	0.062	11.295	Accepted
AI-WB	0.661	0.086	7.697	Accepted
AI-NC	0.519	0.080	6.519	Accepted
NC-WP	0.735	0.067	10.890	Accepted
NC-WB	0.814	0.081	10.016	Accepted

Mediation Analysis

Mediation testing was conducted to explore indirect effects through intermediate variables. The mediation tests showed significant indirect effects in both paths analyzed:

- **AI → NC → WP:** The indirect effect ($a * b$) was calculated as 0.241, with a direct effect (c') of 0.461. This suggests that NC partially mediates the relationship between AI and WP.
- **AI → NC → WB:** The indirect effect was 0.318, with a direct effect (c') of 0.343, showing that NC also partially mediates the relationship between AI and WB.

These results highlight the importance of the NC factor in mediating the relationships between AI and both WP and WB.

Moderation Analysis

Finally, a moderation analysis was performed to investigate whether **Empowering Leadership (ELB)** moderates the relationship between **AI** and **NC**. The results of the **Moderation Test** indicate that the interaction term ($AI * ELB$) is not statistically significant (**p-value = 0.910**). The low coefficient for the interaction term (-0.0164) suggests that Empowering Leadership does not significantly moderate the relationship between AI and NC.

Conclusion and Recommendations

This study contributes to the growing body of research on the socio-psychological impacts of Artificial Intelligence (AI) in organizational contexts by integrating both the Technology Acceptance Model (TAM) and Self-Determination Theory (SDT). The findings indicate that AI integration, when supported by a psychologically empowering work environment, significantly enhances both performance and well-being of leaders and employees. Specifically, perceived competence mediates these positive effects, and empowering leadership further strengthens this relationship—especially for employees who rely on support and autonomy to navigate AI transitions.

From a practical standpoint, organizations should prioritize fostering perceived competence among employees and leaders. This includes regular AI training workshops, establishing clear guidelines for AI tool usage, and creating opportunities for employees to apply new skills. Equally vital is the adoption of empowering leadership practices, such as encouraging innovation, providing necessary resources, and recognizing AI-related accomplishments.

Recommendations:

- Implement AI-Specific Training Programs to boost perceived competence across organizational levels.
- Develop Leadership Development Initiatives focused on empowering styles to facilitate smoother AI adoption.
- Customize AI Integration Plans based on employee roles and experience to reduce stress and increase confidence.
- Promote Psychological Safety and Well-being by openly addressing anxieties related to AI disruption.
- Continuously Evaluate AI Impact through performance and well-being metrics to ensure sustainable growth.

Limitations and Future Research

Despite offering valuable insights, this study has several limitations. First, it relies on self-reported data, which can introduce bias due to social desirability or subjective interpretation. Second, the sample was limited to the technology sector in Pakistan, which restricts generalizability across industries or countries with different digital maturity levels. Third, the cross-sectional design prevents the study from establishing causal relationships.

Future research should consider:

- Longitudinal studies to explore changes in AI impact over time.
- Comparative studies across industries and regions to identify sector-specific challenges and strengths.
- Qualitative approaches like interviews and focus groups to delve deeper into employee and leader experiences.
- Exploring additional mediators and moderators (e.g., organizational culture, team dynamics, or job complexity).
- Examining AI's impact on other psychological outcomes, such as job satisfaction, learning agility, or innovation.

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